

BHAVISH RAI B

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🌐 [Bhavish Rai B](#) 🔄 [braib](#) 🌐 [braib.github.io](#)



Objective

Programming and building robots that can operate effectively in complex and real-world applications.

Education

Bachelor of Engineering in Robotics and Automation <i>Sahyadri College of Engineering and Management, Mangaluru</i>	2021 - 2025 CGPA: 8.31/10.0
Pre-University Education (PCMCs) <i>Sri Rama Pre-University College, Kalladka</i>	2019 - 2021 Percentage: 86%
Secondary Education (SSLC) <i>Govt. High School Beliyurukatte, Puttur</i>	2019 Percentage: 84.64%

Professional Experience and Projects

Research Intern (Jan - May 2025), Program Assistant - I (June 2025 - Present)

Indian Institute of Technology, Gandhinagar

- Contributed to design and code for ROS2 architecture of Collaborative Multirobot System(CoMuRoS) [\[Website\]](#)
- Contributed to finetuning of prompts for VLM and LLM for event detection and Task Manager of CoMuRoS. [\[Video\]](#)
- Contributed develop a Navigation Manager that converts high-level natural language tasks into metric goal poses and implemented path planners for both single-robot and multi-robot navigation in heterogeneous robot systems. [\[Video\]](#)
- Vision+PID based Controllers for mobile robots, implemented control for holonomic robot and kanayama controller for differential drive robot. [\[Video\]](#)
- Curated Dataset of Scenario Configurations and Tasks for testing generalization of CoMuRoS across Scenarios, Robots, Tasks and LLMs.
- Implemented ChoiRbot library for formation control of mobile robots on Turtlebot3 Hardware platform. [\[Video\]](#)
- Implemented position control on actual Unitree Go2 robot. [\[Video\]](#)
- Developed multiple Gazebo environments like Hospital, Disaster Relief and Restaurant for testing of CoMuRoS. [\[Video\]](#)
- Fabricated 3 Lerobot Arms. Performed data collection, finetuning of Nvidia Groot VLA and successful testing of pick and place of objects using VLA. [\[Video\]](#)
- Built bi-manual social tutorbot from scratch using Lerobot and LLMs, this robot could communicate through voice commands and guided students in assembly tasks using the CoMuRoS architecture as backbone. [\[Video\]](#)
- Contributed work to a talk in ROSCON 2025, multiple research publications and one patent application. [\[Video\]](#)
- Implemented pick and place on X-arm Manipulator using MoveIt. [\[Video\]](#)
- Implemented Nav2 on Turtlebot3.
- Implemented a Navigation Manager module that combines Task Planning of CoMuRoS with Path Planning.

Research Intern - Robotic Puppy Dog Project (October 2023 - November 2023)

Technical Career Education, Adyar, Mangaluru

- Designed and developed a quadruped robotic platform with actuated joints and integrated sensors
- Prototyped the physical leg fixture of the robot dog with servo motors

Technical Projects

CoMuRoS: Collaborative Multi-Robot System

IIT Gandhinagar

- Developed a generalizable hierarchical architecture for LLM-driven task planning and event-driven replanning in heterogeneous multi-robot teams, unifying centralized deliberation with decentralized execution
- Implemented Task Manager LLM for natural-language goal interpretation and dynamic task allocation using static rules and contextual information (task history, robot status, events) [\[Video\]](#)
- Designed local robot LLMs that compose executable Python code from primitive ROS2 skills, with onboard VLM-based perception for continuous event monitoring and classification [\[Video\]](#)
- Achieved 90% collaborative object recovery success (9/10), 100% coordinated transport (8/8), and 100% human-assisted recovery (5/5) in hardware experiments
- Demonstrated autonomous recovery from disruptive events, filtering of irrelevant distractions, and emergent human-robot cooperation through runtime replanning [\[Video\]](#)
- Developed using ROS2 Humble with Python, simulated in Gazebo Classic and Ignition Fortress [\[Video\]](#)

Integrated Task and Kinodynamic Path Planning for Heterogeneous Multi-Robot Formations

IIT Gandhinagar

- Developed a Navigation Manager that converts high-level natural language tasks into metric goal poses understandable by path planners.
- Implemented path planners for both single-robot and multi-robot navigation in heterogeneous robot systems. [\[Video\]](#)

An Agentic AI Architecture for Robot-Assisted Hands-on Learning.

IIT Gandhinagar

- Developed an LLM-driven Bimanual TutorBot capable of interacting with users through speech and facial expressions [\[Video\]](#)
- Integrated two LeRobot arms as bimanual manipulators for interactive tutoring tasks
- Used a display screen to show robot's facial expressions for improved human-robot interaction
- Implemented CoMuRoS architecture by treating the two robotic arms as a multi-robot system
- Enabled real-time assistance for students performing assembly tasks
- Designed voice-based interaction using microphone input for student queries
- Implemented Socratic-style guidance through verbal responses and action-based demonstrations
- Enabled object picking and visual hint demonstration to assist task completion

3-DOF Robotic Arm Manipulator

- Designed a 3-degree-of-freedom robotic arm using SolidWorks
- Implemented forward and inverse kinematics for precision control and trajectory planning
- Developed control algorithms for pick-and-place automation tasks

Poultry Patrol Robot

- A robot that detects and marks diseased chickens in poultry farms using computer vision and ML [\[Video\]](#)
- Designed a differential drive mobile robot in SolidWorks (2D/3D) for autonomous poultry farm surveillance, fabricated using aluminum sheet metal, 3D-printed PLA parts, and black acrylic panels
- Integrated Raspberry Pi 5 with YOLOv5 for real-time diseased chicken detection via webcam, and ESP32 for distributed sensor data acquisition and wireless telemetry
- Implemented motor control via L298N drivers and a servo-actuated color sprayer mechanism to autonomously mark identified diseased birds
- Built on ROS2 with dedicated nodes for image capture, ML inference, and motor control, enabling autonomous navigation and obstacle avoidance in farm environments
- [\[Report\]](#)

Automated Home Security System

- Developed IoT-based surveillance system with computer vision for intruder detection
- Integrated Telegram bot API for real-time alerts and remote monitoring capabilities
- Implemented motion detection and image capture using Raspberry Pi and camera module
- [\[Report\]](#)

Automated Farm Water Pump System

- Designed an IoT-based smart irrigation system for agricultural use, automating water pump control based on real-time soil moisture data

- Integrated soil moisture sensor and motor driver with ESP32, streaming live sensor data to Adafruit IO dashboard for remote monitoring
- Implemented event-driven email notifications via IFTTT, alerting farmers instantly on every pump state change without manual checking

Design and Analysis of Helical Spring

- Designed a helical spring ($d=14\text{mm}$, $D=70\text{mm}$, 10 active coils) in SolidWorks and performed nonlinear FEA simulation for shear stress and deflection under a 3768.7N load
- Validated FEA results against hand-calculated theoretical values using Wahl's correction factor, achieving 1.14% error in deflection (35mm vs 35.4mm)
- Developed a Python script to parametrically analyze the effect of mean diameter on induced shear stress, plotting results across a range of 55–85mm
- [\[Report\]](#)

Publications

CoMuRoS: A LLM-Driven Hierarchical Architecture for Adaptive Multi-Robot Collaboration. [\[Paper\]](#) [\[Website\]](#)

Suraj Borate, **Bhavish Rai B**, Vipul Pardeshi and Madhu Vadali. IROS 2025 Workshop - LeaPRiDE: Learning, Planning, and Reasoning in Dynamic Environments, Hangzhou, China, 2025.

LLM-Based Generalizable Hierarchical Task Planning and Execution for Heterogeneous Robot Teams with Event-Driven Replanning. [\[Paper\]](#) [\[Website\]](#)

Suraj Borate, **Bhavish Rai B**, Vipul Pardeshi and Madhu Vadali. ArXiv, 2025. Under Review at Frontiers in Robotics and AI.

Application of LLM Driven Robot Architecture for the Study of Robot Interventions in Making Task.

Suraj Borate, **Bhavish Rai B***, Megha Bansal*, Harshil Shafi, Saloni Shinde, Aditi Kothiyal, and Madhu Vadali. Submitted to IEEE Transactions on Learning Technologies, 2026.

Integrated Task and Kinodynamic Path Planning for Heterogeneous Multi-Robot Formations

Suraj Borate, **Bhavish Rai B**, and Madhu Vadali.

Accepted for ICRA 2026 Workshop Scaling Compositional Intelligence for Multi-Agent Robotic Systems (SCI-MARS).

IEEE Robotics and Automation Letters (RA-L). *[Under preparation]*

Patents

A Multi-Robot Coordination System for Task Planning and Execution

Patent Applied

Inventors: Suraj Borate, Madhu Vadali, and **Bhavish Rai B**

Filing Date: 07.05.2026

Technical Skills

Programming Languages: Python, C++, C, I can adapt to new language quickly.

Robotics & Simulation: ROS2 Humble, Gazebo Classic, Ignition Fortress, RVIZ

CAD & Design: SolidWorks, Fusion 360

Embedded Systems: Arduino, ESP32, ESP8266, Raspberry Pi, Jetson Nano Orin NX

Development Tools: Git, GitHub, VS Code, Linux (Ubuntu 22.04)

Prototyping: 3D Printing, Laser Cutting, Soldering

Soft Skills: Problem Solving, Team Collaboration, Project Management, Rapid Prototyping, Event Organization

Robot Platforms Experience

Custom Platforms: Poultry Robot

Quadruped Robots: Unitree GO2 (Edu)

Wheeled Mobile Robots: TurtleBot4, TurtleBot3 Burger, TurtleBot3 Waffle Pi, LeKiwi

Manipulators: SO101 Arm, TurtleBot3 Waffle Pi with Manipulator, Open Manipulator X

Achievements & Leadership

SAE AEROTHON 2024: Secured All India Rank 22 in national-level aerospace competition

Mattarme Robo Race: First Place in autonomous robot racing competition

Aerophilia 2023: Organized techno-cultural fest as core team member (December 2023)

Team Challengers Club: Contributed to team robotics projects while mentoring junior members across a 3-year period (February 2022 - January 2025)

Certifications

ROS2 for Beginners Level 1 & 3 - ROS2 fundamentals to advanced robotics (April 2024, August 2024)

[\[Certificate 1\]](#) [\[Certificate 2\]](#)

OpenCV Bootcamp - Computer vision and image processing fundamentals (May 2024) [\[Certificate\]](#)